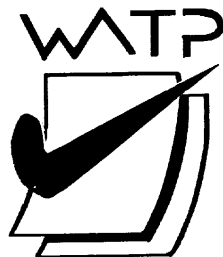


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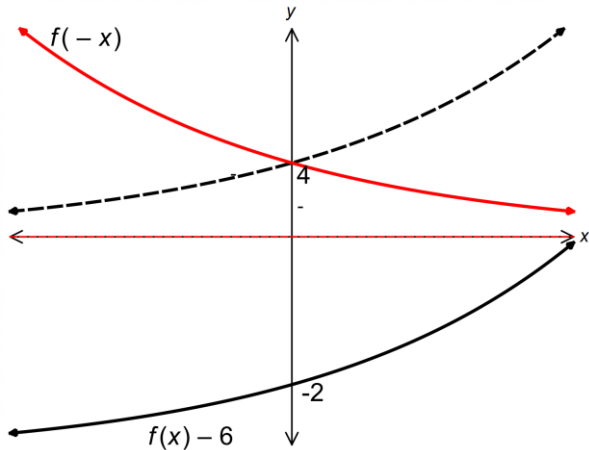
SEMESTER TWO

MATHEMATICS METHODS UNITS 1 & 2

2019

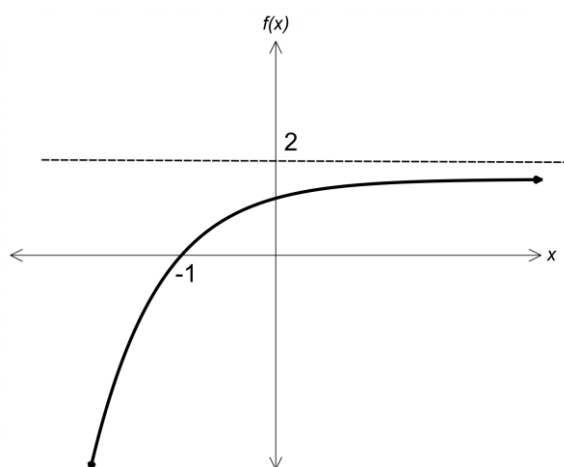
SOLUTIONS

Calculator-free Solutions

1. (a) $2^{-2} \times a^{-6} \times b^2$ ✓
 $= \frac{b^2}{4a^6}$ ✓
- (b) $\frac{3^{-3} \times x^9 \times y^{-6}}{9^{-2} \times x^4 \times y^{-4}}$ ✓
 $= \frac{81 \times x^5}{27y^2}$ ✓
 $= \frac{3x^5}{y^2}$ ✓ [5]
2. (a) $2^{2-2x} = 2^{3-3x}$ ✓
 $\therefore 2 - 2x = 3 - 3x \rightarrow x = 1$ ✓
- (b) $3^{x^2+1} = 3^{x+3}$ ✓
 $\therefore x^2 + 1 = x + 3$ ✓
 $\therefore x^2 - x - 2 = 0 \rightarrow (x-2)(x+1) = 0$ ✓
 $\therefore x = 2 \text{ or } x = -1$ ✓ [6]
3. (a) (i) $m = 4$ ✓
(ii) $f(-2) = 4a^{-2} = \frac{4}{a^2}$ ✓
- (b)  ✓✓✓✓ [6]
4. (a) (i) $\frac{dy}{dx} = 3x^3 + 5x^4$ ✓✓
(ii) $f(x) = \frac{x}{3} + \frac{2x^2}{\pi}$
 $\therefore f'(x) = \frac{1}{3} + \frac{4}{\pi}x$ ✓✓
- (b) $-3x^2$ ✓ [5]

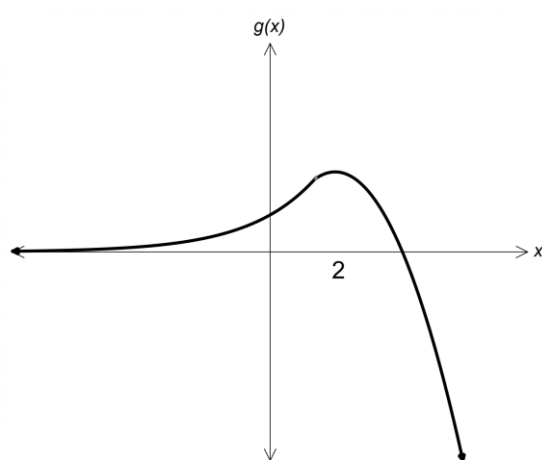
5. (a) (i) $y(1) = -4$ and $y(3) = -18$ ✓
 $\therefore$ Av rate of change = $\frac{-18 - (-4)}{2} = -7$ ✓
(ii) $y' = 3t^2 - 10t$ ✓
 $\therefore y'(2) = -8$ ✓
(b) $\delta y = (3t^2 - 10t) \times 2$ ✓
 $\therefore \delta y = (-7) \times 2 = -14$ ✓
(c) $\frac{dy}{dt} = -2 \rightarrow y = -8t + c$ ✓
 $\therefore (2, -12) \rightarrow -12 = -16 + c \rightarrow c = 4$ ✓
 $\therefore y = -8t + 4$ ✓ [9]
6. (a) (i) $t = 1.5$ ✓
(ii) $t = 0.5$ ✓
(b) $x(t) = a(t - 0.5)^2 + 2.5$ ✓
 $(0, 2) \rightarrow 2 = a(0.25) + 2.5 \rightarrow a = -2$ ✓
 $\therefore x(t) = -2(t - 0.5)^2 + 2.5$ ✓ [4]
7. (a) $\frac{dy}{dx} = 2x - \pi^2 + \frac{3}{5}x^2$
 $\therefore y = x^2 - \pi^2 x + \frac{1}{5}x^3 + c$ ✓✓
(b) $f'(n) = \frac{1}{3}n^2 - \frac{4}{3}n$ ✓
 $\therefore f(n) = \frac{1}{9}n^3 - \frac{2}{3}n^2 + c$ ✓✓ [5]
8. (a) $T_{n+1} = T_n + 10$ where $T_1 = 100$ ✓
(b) $T_n = 100 + (n - 1)(10) = 10n + 90$ ✓
(c) $T_8 = 10(8) + 90 = 170$ km ✓✓
(d) $S_{12} = 6(200 + 110) = 6(310)$ ✓
 $\therefore = 1860$ km ✓ [6]

9. (a)



✓✓✓

(b)

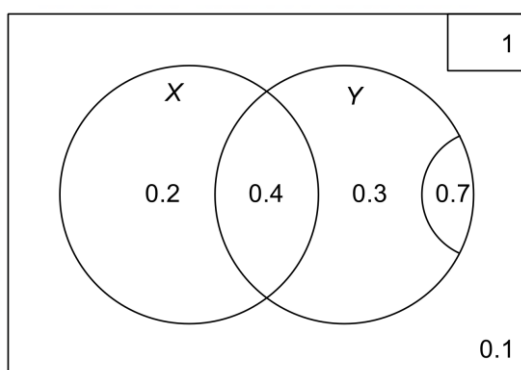


✓✓✓

[6]

Calculator–Assumed Solutions

10. (a)



(b) (i) 0.6 ✓✓

(ii) $\frac{0.6}{0.9} = \frac{2}{3}$ ✓

(c) $P(X|Y) = \frac{0.4}{0.7}$ and $P(X) = 0.6$ ✓✓

∴ not independent since $P(X) \neq P(X|Y)$ ✓

or $P(X) \times P(Y) \neq P(X \cap Y)$

[7]

11. (a) $|r| < 1$ ✓

(b) $ar = 20$ and $\frac{a}{1-r} = 125$ ✓

∴ $a = 25$ and $r = 0.8$ or $a = 100$ and $r = 0.2$ ✓✓

(c) $S_{10} = \frac{25(1-0.8^{10})}{1-0.8} = 111.6$ ✓

or $S_{10} = \frac{100(1-0.2^{10})}{1-0.2} = 125.0$ ✓

[6]

12. (a) (i) $(c, 0), (g, 0), (k, 0)$ ✓✓

(ii) $(e, f), (i, j)$ ✓✓

(iii) (a, b) ✓

(b) $a \leq x < e, g < x < i, k < x \leq m$ ✓✓

(c) $e < x < g, i < x < k$ ✓✓

(d) $k = 2$ ✓

[10]

13. (a) 0.98 means reducing by 2% ✓

(b) $N(12) = 3500(0.98)^{12} = 2747$ rats ✓✓

(c) $3500(0.98)^t = 2000$ ✓

∴ $t = 27.7 \rightarrow$ April 2021 ✓

(d) $3200 = 2000r^{12} \rightarrow r = 1.04$ ✓

∴ $R(t) = 2000(1.04)^t$ ✓

(e) $3500(0.98)^t = 2000(1.04)^t$ ✓✓

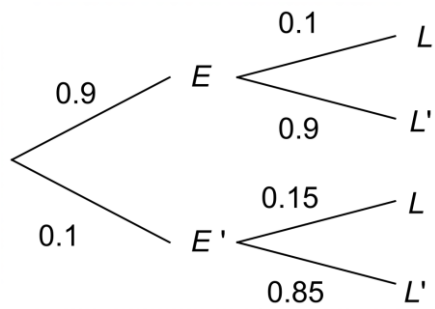
∴ $t = 9.4$

∴ October 2019 ✓

[10]

14. (a) 10L ✓
 (b) $W'(t) = 0.8t^3 - 1.8t^2 - t$ ✓
 $\therefore 0.8t^3 - 1.8t^2 - t = 0$ when $t = 2.7$ ✓
 $\therefore$ During the third minute. ✓
 (c) (i) $W(2.7) = 5.2$ L ✓
 (ii) $W(4) = 14.8$ L ✓ [6]
15. (a) (i) -15 m ✓
 (ii) $v(t) = t^2 - 2t - 4$ ✓✓
 (b) $v(3) = -1$ ✓
 $\therefore$ Speed = 1 m/s ✓
 (c) At rest when $v(t) = 0$
 $\therefore t^2 - 2t - 4 = 0$ ✓
 $\therefore t = 3.2$ s ✓
 (d) $x(0) = -3$
 $x(3.2) = -15.12$ ✓
 $x(5) = -6.33$ ✓
 $\therefore$ Distance travelled = $12.12 + 8.79 = 20.91$ m ✓ [10]
16. (a) $c = 5$ ✓
 (b) $(-2, -3) \rightarrow -3 = -8 + 4a - 2b + 5$
 $\therefore 4a - 2b = 0$ ✓
 $\frac{dy}{dx} = 3x^2 + 2ax + b = 0$ when $x = -2$
 $\therefore 4a - b = 12$ ✓
 $\therefore a = 6$ and $b = 12$ ✓✓
 (c) $y = x^3 + 6x^2 + 12x + 5$
 $\therefore x^3 + 6x^2 + 12x + 5 = 0$ when $x = -0.558$ ✓✓ [7]
17. (a) $\frac{dy}{dx} = 15x^4 - 30ax^2 + 15a^2$
 $\therefore y = \frac{15x^5}{5} - \frac{30ax^3}{3} + 15a^2x + c$ ✓✓
 $\therefore (0, 0) \rightarrow c = 0$ ✓
 $\therefore y = 3x^5 - 10ax^3 + 15a^2x$
 (b) $\frac{dy}{dx} = 15(x^2 - a)^2 = 0$ when stationary ✓
 $\therefore (x^2 - a)^2 = 0$
 $\therefore x^2 = a$ ✓
 Since two stationary points $a > 0$ ✓ [6]

18. (a)



(b) (i) $P(L) = 0.9 \times 0.1 + 0.1 \times 0.15 = 0.105$

(ii) $P(L'|E') = 0.85$

(iii) $P(E|L) = \frac{0.9 \times 0.1}{0.105}$
 $= 0.857$

✓✓

✓✓

✓

✓

✓

[7]

19. $a = 0.5$

$$1.5T = 2\pi \rightarrow T = \frac{4\pi}{3}$$

$$\therefore b = \frac{3}{2}$$

$$c = \frac{\pi}{4}$$

$$d = -2$$

$$e = -1$$

✓

✓

✓

✓

✓

✓

[6]

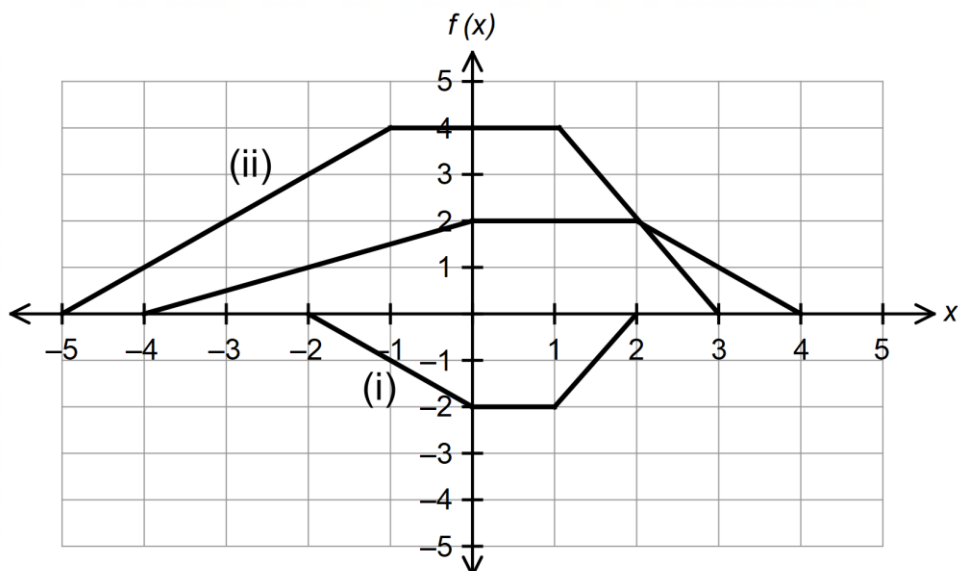
20. (a) (i) $x = \pm 4$

(ii) 2

(b)

✓

✓



✓✓✓✓

[6]

21. (a) $g'(x) = 4x^3 - 16x$ ✓
 $g'(1) = -12$ ✓
 $\therefore y = -12x + c$
 $(1, 9) \rightarrow 9 = -12(1) + c \rightarrow c = 21$ ✓
 $\therefore y = -12x + 21$ ✓
 (b) $(1.45, 3.61)$ and $(-3.45, 62.4)$ ✓✓ [6]

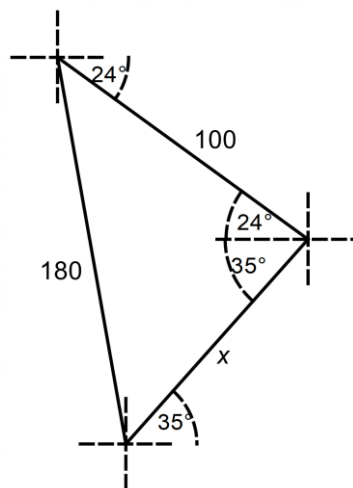
22. (a)
$$\frac{(\sin a \cos 45^\circ + \sin 45^\circ \cos a)(\cos a \cos 45^\circ - \sin 45^\circ \sin a)}{(\sin a - \cos a)(\sin a + \cos a)}$$
 ✓

$$= \frac{\left(\frac{\sqrt{2}}{2} \sin a + \frac{\sqrt{2}}{2} \cos a\right)\left(\frac{\sqrt{2}}{2} \cos a - \frac{\sqrt{2}}{2} \sin a\right)}{(\sin a - \cos a)(\sin a + \cos a)}$$
 ✓

$$= \frac{\frac{\sqrt{2}}{2}(\sin a + \cos a) \frac{\sqrt{2}}{2}(\cos a - \sin a)}{(\sin a - \cos a)(\sin a + \cos a)}$$
 ✓

$$= \left(\frac{2}{4}\right)(-1) = -\frac{1}{2}$$
 ✓

(b)



$$180^2 = 100^2 + x^2 - 2x(100)\cos 59^\circ$$
 ✓
 $\therefore x = 210 \text{ km}$ ✓✓ [8]

23.
$$\left(\sin x + \frac{1}{\sin x}\right)^3$$

$$= \sin^3 x + 3(\sin^2 x)\left(\frac{1}{\sin x}\right) + 3(\sin x)\left(\frac{1}{\sin x}\right)^2 + \left(\frac{1}{\sin x}\right)^3$$
 ✓✓

$$= \frac{1}{\sin^3 x}(\sin^6 x + 3\sin^4 x + 3\sin^2 x + 1)$$
 ✓ [3]